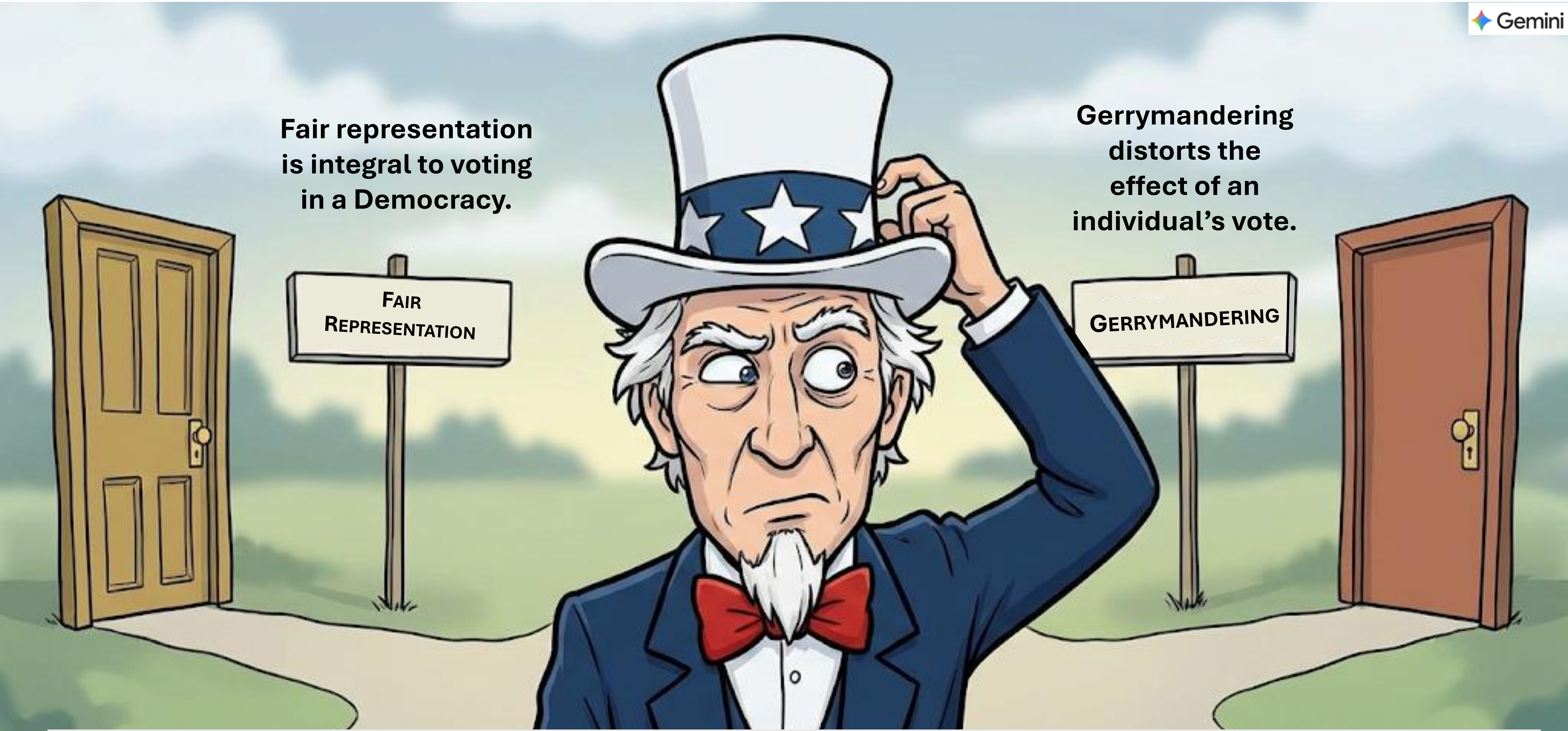
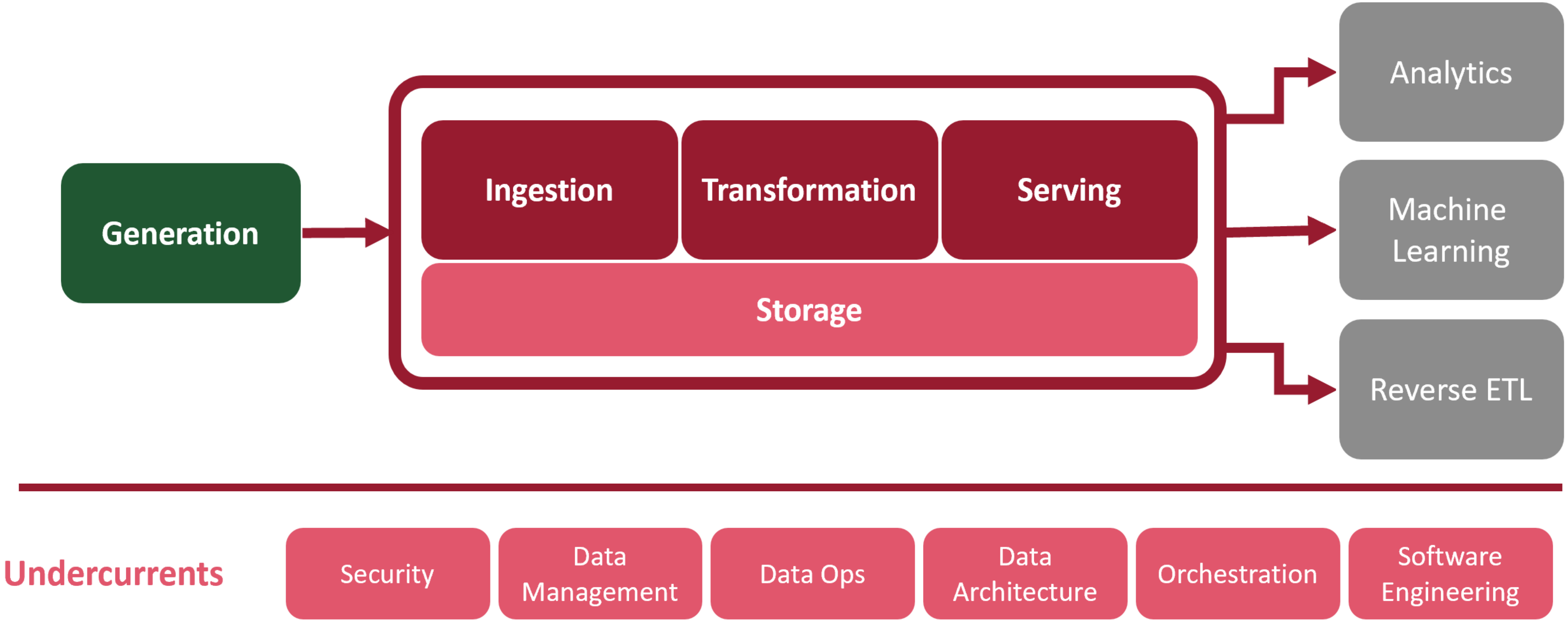


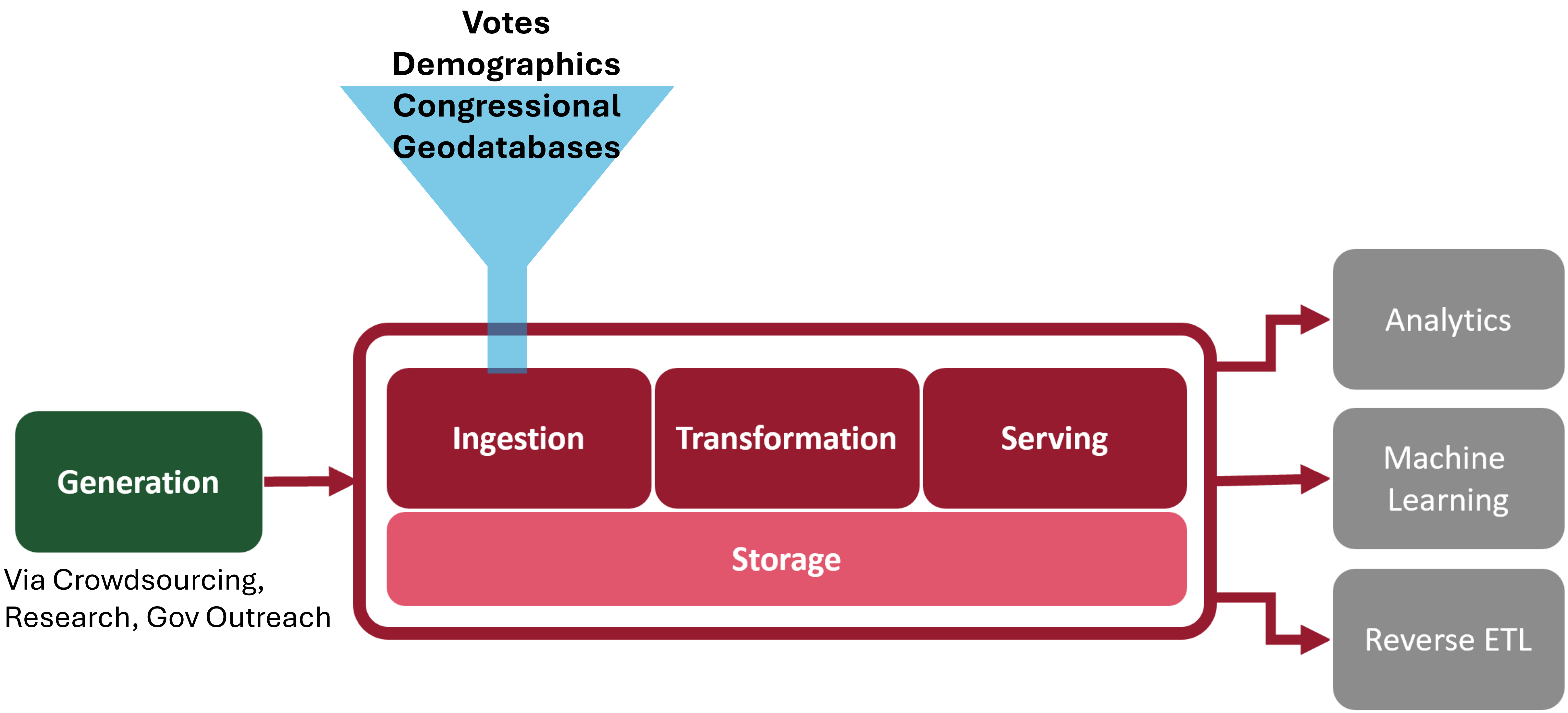

# Identifying Partisan Gerrymandering Risk in U.S. Congressional Districts

# Following The DE Lifecycle



# Data Sources

| Data | Source | Summary | Documentation Process | Accessability |
|---|---|---|---|---|
| Voter Data | **Open Elections**<br>https://openelections.net/ | Precinct-level voter data | Crowdsourced | GitHub (CSVs) |
| Voter Data | **Univ. of Florida Election Lab**<br>https://election.lab.ufl.edu/precinct-data/ | Voter turnout, counts, early voting | Faculty/student generated | Website (Shapefile) |
| Demographics | **Census Bureau – American Community Survey**<br>https://www.census.gov/data/developers/data-sets/acs-5year.html | Population, Race, Sex, Income, etc. | Federal government (via US Census) | API |
| Geographic | **Census Bureau – REST Map Services**<br>https://tigerweb.geo.census.gov/tigerwebmain/TIGERweb_restmapservice.html | Land use, traits | Federal government | API |


Votes
Demographics
Congressional
Geodatabases
Generation
Via Crowdsourcing, Research, Gov Outreach
Ingestion
Transformation
Serving
Storage
Analytics
Machine Learning
Reverse ETL

# Transformation

## Filtering and standardizing Open Elections for Congressional data

## Calculating voting power, partisan symmetry

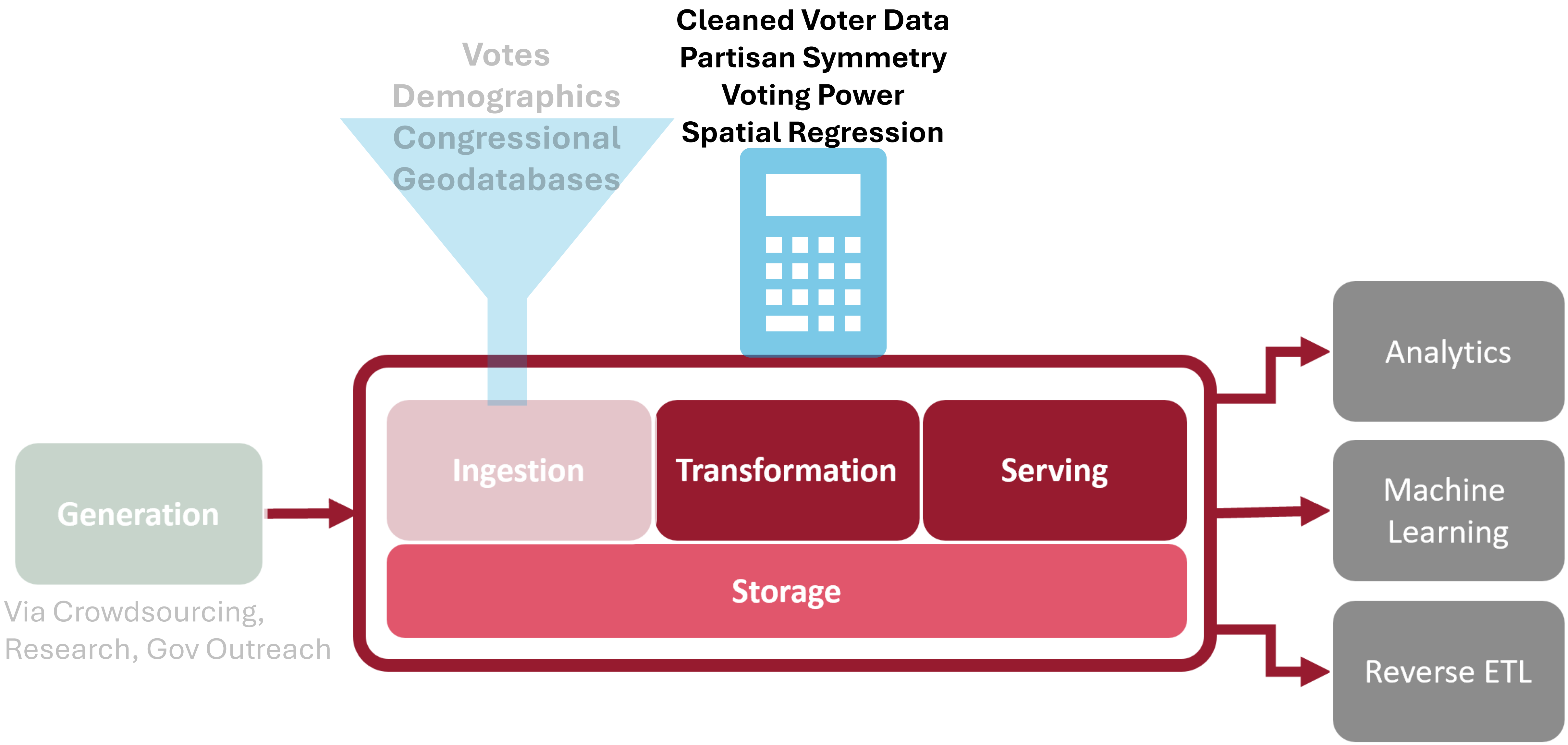
- Seats-Votes Curve, Efficiency Gap, Mean-Median Score

## Demographics

- Distribution of populations, populations from precinct to Congressional District

## Geospatial calculations

- Spatial Regression – geography as a feature
- Reock Score to measure compactness
- Boundaries (County, City, Precinct) present within a district, crossing districts
- Prior line control (partisan)


Cleaned Voter Data
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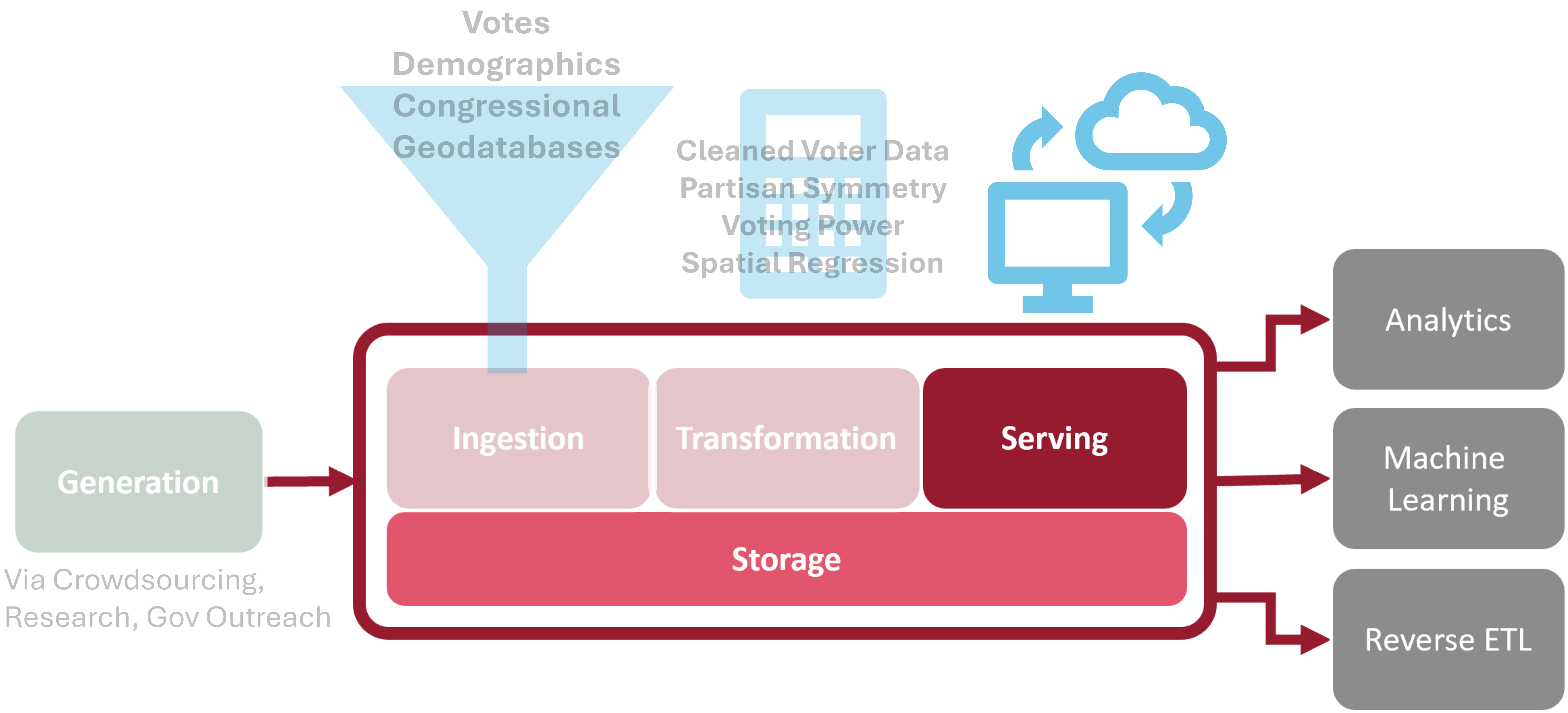
# Orchestration, Storage, Service

- Cloud storage for source data
- Transform within cloud
- Serve transformed and raw data via API to users for analysis



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